

critical communications, and essential services remain protected, regardless of technological advances. As organisations and governments move towards adopting these standards, quantum security algorithms are becoming an indispensable cornerstone for the future of data protection, inspiring confidence in the continued safety of digital interactions and transactions.

Post-Quantum Cryptography (PQC) represents a pivotal advancement in the ongoing evolution of cryptographic security, emerging in direct response to the accelerating development of quantum computing technologies. The primary aim of PQC is to develop and standardise cryptographic algorithms that can withstand the unprecedented computational capabilities of quantum machines. As traditional public key cryptosystems, which currently underpin secure communications and digital trust, face obsolescence in the quantum era, the adoption of robust PQC algorithms is essential for maintaining the confidentiality, authenticity, and integrity of sensitive information.

Quantum security algorithms — the foundation of data protection in the future

Lattice-based cryptography

algorithms: These draw upon advanced branches of mathematics to create formidable barriers against even the most sophisticated quantum attacks. Lattice-based cryptography leverages the inherent complexity of lattice problems, which quantum computers find exceptionally challenging to break. Similarly, hash-based cryptography relies on the irreversible nature of cryptographic hash functions, making it difficult for adversaries to reconstruct original data. Code-based and multivariate polynomial-based schemes further expand this repertoire, offering a variety of mechanisms that collectively strengthen the protective framework around sensitive digital assets.

Shor's algorithm: The critical security threat posed by quantum computing arises from Shor's algorithm, a groundbreaking quantum algorithm devised by mathematician Peter Shor. Shor's algorithm efficiently factors large integers and computes discrete logarithms, tasks that are prohibitively time-consuming for classical computers. This efficiency directly undermines the security basis of widely used public key systems such as RSA, which relies on the difficulty of factoring large numbers, and Elliptic Curve Cryptography (ECC), which depends on the hardness of the discrete logarithm problem over elliptic curves. In theory, a sufficiently powerful quantum computer running Shor's algorithm could break these systems in a matter of hours, rendering encrypted communications, digital signatures, and key exchanges vulnerable to interception and forgery. This imminent risk has accelerated the global search for quantum-resistant alternatives to current cryptographic standards.

CRYSTALS-Kyber algorithm: PQC algorithms based on lattice problems are believed to be resistant to both classical and quantum attacks. Among these, CRYSTALS-Kyber has gained significant attention as a key encapsulation mechanism (KEM) designed for secure key exchange. This algorithm utilises the hardness of the Learning With Errors (LWE) problem, a mathematical challenge for which no efficient quantum solution is known. In practice, when two parties wish to establish a shared secret over an insecure channel, CRYSTALS-Kyber enables them to exchange messages in such a way that an eavesdropper, even with access to a quantum computer, cannot feasibly compute the secret key.

For instance, in a secure messaging application, Alice can use Kyber to generate a public key and share it with Bob. Bob then uses this public key to encapsulate a secret, sending the resulting ciphertext back to Alice. Only Alice, with her private key, can decapsulate the ciphertext and retrieve the shared secret, ensuring both confidentiality and forward secrecy in the presence of quantum threats.

CRYSTALS-Dilithium algorithm:

Complementing key exchange mechanisms, secure digital signatures are indispensable for authentication and non-repudiation in digital transactions. The CRYSTALS-Dilithium algorithm, also founded on lattice-based cryptography, establishes a quantum-resistant digital signature scheme. Built on the Module-LWE and Module-SIS problems, it offers strong security guarantees and efficient implementation. In a typical scenario, an organisation may use Dilithium to sign software updates before distribution. The developer generates a signature using their private key, which is then attached to the update package. Recipients, such as end users or automated systems, can verify the authenticity of the update using the corresponding public key, confident that even a quantum-enabled adversary cannot forge signatures or impersonate the sender. This ensures the integrity and provenance of critical software, protecting against tampering and supply chain attacks.

As quantum computing continues to advance, the necessity for robust post-quantum cryptography solutions becomes ever more apparent. Understanding the distinguishing features, strengths, and typical applications of various algorithms is essential for organisations seeking to safeguard their digital assets against future quantum-enabled threats (see Table).